Fiber bundle

base manifold M

fiber F (manifold)

bundle: another manifold E s.t. E locally looks like $M \times F$

attaching a copy of F at each point in M

$F = (-1,1)$ $\quad M = S^1$ $\quad E = S^1 \times (-1,1)$ cylinder

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Def E, M, F smooth manifolds. E is a fibre bundle with base M and fibre F if

- there is a smooth surjective map $\pi : E \to M$ (bundle projection)
- E locally looks like $M \times F$. Specifically for each $a \in M$, there is an open set $U \subseteq M$ and a diffeomorphism $\phi : \pi^{-1}(U) \underset{\text{open}}{\subseteq} E \to U \times F$

  such that $\underline{\text{proj}_1} \circ \phi = \pi$

  $\hookrightarrow$ $\text{proj}_1 : U \times F \to U$

  if $x \in \underbrace{\pi^{-1}(a)}$ then $\phi(x) \in \{a\} \times F$

  $\hookrightarrow$ Fibre attached to point $a \in M$

(U, ϕ) is called a local trivialisation

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for each $a \in M$, $F_a = \pi^{-1}(a)$ copy of F at point a

<u>note:</u> $F_a \cap F_b = \emptyset$ if $a \neq b$

$$\Rightarrow \boxed{E = \bigcup_{a \in M} F_a}$$ (disjoint union)

the map $x \in F_a \mapsto \phi(x) \in \{a\} \times F$ diffeomorphism

$$F_a \cong \{a\} \times F \cong F$$

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the open sets U in the trivialisations are <u>not</u> the ones from the charts of M.

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ex <u>Trivial bundle</u> $E = M \times F$ $\quad U = M$

$\pi : (a,f) \in M \times F \to a \in M$ $\quad \phi : \mathrm{id}_E$

more generally, E is trivial if it has a global trivialisation.

<u>Tangent bundle</u> $E = TM = \bigcup_{a \in M} \overbrace{\{a\} \times T_a M}^{F_a}$ $\quad \pi : (a,s) \in TM \mapsto a \in M$

$F = \mathbb{R}^n \quad n = \dim(M)$

if $(U, (x^1, \dots, x^n))$ coordinate chart for M

$$\phi : \left(a, v^i \frac{\partial}{\partial x^i}\Big|_a\right) \in \pi^{-1}(U) = (a, v^1, v^2, \dots, v^n) \in U \times \mathbb{R}^n \qquad a \in U \subseteq M$$

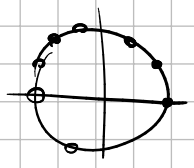
coordinate basis for $T_a M$

for $M = S^1$ $\quad TS^1 \cong S^1 \times \mathbb{R}$

<u>Tautological line bundles</u> $M = P^1(\mathbb{R}) \cong S^1$

$P^1(\mathbb{R}) = \{ [(x,y)] \mid (x,y) \in \mathbb{R}^2 \setminus \{0\} \}$ $\quad [(x,y)] = \{ (\lambda x, \lambda y) \mid \lambda \in \mathbb{R} \} \cong \mathbb{R}$

$E = \{ ([x,y], s) \mid (x,y) \in \mathbb{R}^2 \setminus \{0\},\ s \in [(x,y)] \}$



$\pi : ([x,y], s) \in E \mapsto [x,y] \in P^1(\mathbb{R})$

$F = \mathbb{R}$ $\quad (u,v) \cdot (1, y/x)$

$U_1 = \{ [x,y] \mid x \neq 0 \}$

$U_2 = \{ [x,y] \mid y \neq 0 \}$

$\phi_1 : ([x,y], v, s) \in \pi^{-1}(U_1) \mapsto \left([x,y], v + s\frac{y}{x}\right) \in U_1 \times \mathbb{R}$

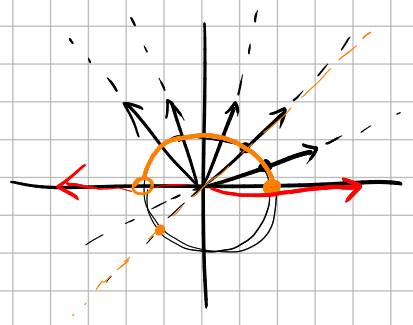
$\phi_2 : ([x,y], v, s) \in \pi^{-1}(U_2) \mapsto \left([x,y], \frac{vx}{y} + v\right) \in U_2 \in \mathbb{R}$

$\phi_1^{-1} : ([x,y], \lambda) \in U_1 \times \mathbb{R} \mapsto \left([x,y], \lambda, \frac{\lambda y}{x}\right)$

$\in [x,y]$

$\phi_2^{-1} : ([x,y], \lambda) \in U_1 \times \mathbb{R} \mapsto \left([x,y], \frac{\lambda y}{x}, \lambda\right)$

$E \cong$ Mobius band $\qquad (P^1(\mathbb{R}) \cong S^1)$



$\mathbb{R}^2$

all examples of vector bundles

in general, F may have extra structure (vector space or something else)

Def: a real vector bundle of rank n is a fibre bundle $\pi : E \to M$ with fibre $F = \mathbb{R}^n$ such that the fibres F_a are real vector space and

$x \in F_a \mapsto \phi(x) \in \underbrace{\{a\} \times \mathbb{R}^n}$ are linear

$\hookrightarrow$ vector space with operations

$\alpha(a, v) + \beta(a, w) = (a, \alpha v + \beta w)$

Alternative def: A real vector bundle of rank n is a fibre bundle $\pi : E \to M$ with fibre $\mathbb{R}^n$ s.t. if $\{(U_i, \phi_i)\}_{i \in I}$ are local trivialisations then the transition functions $\phi_{ij} := \phi_i \circ \phi_j^{-1} : (U_i \cap U_j) \times \mathbb{R}^n \to (U_i \cap U_j) \times \mathbb{R}^n$ are such that

$(a, v) \in \{a\} \times \mathbb{R}^n \mapsto \phi_{ij}(a, v) \in \{a\} \times \mathbb{R}^n \qquad a \in U_i \cap U_j$

$\to$ are isomorphisms

→ can then define vector space structure on F_a as follows

$$x \in F_a \mapsto \phi_i(x) \in \{a\} \times F \qquad a \in U_i$$

$$\alpha x + \beta y = \phi_i^{-1}\left(\alpha \phi_i(x) + \beta \phi_i(y)\right)$$

← independent from choice of trivialisation

if $a \in U_i \cap U_j$

$\phi_i \circ \phi_j^{-1}$

$$\phi_j^{-1}\left(\alpha \phi_j(x) + \beta \phi_j(y)\right) = \phi_i^{-1} \circ \phi_{ij}\left(\alpha \phi_j(x) + \beta \phi_j(y)\right)$$

$$= \phi_i^{-1}\Big(\alpha \underbrace{\phi_{ij}(\phi_j(x))}_{\phi_i(x)} + \beta \underbrace{\phi_{ij}(\phi_j(y))}_{\phi_i(y)}\Big)$$

$$\begin{tikzcd}
\pi^{-1}(U) \subseteq E \arrow[r, "\phi"] \arrow[d, "\pi"'] & \boxed{U \times F} \arrow[d, "\mathrm{proj}_1"] \\
U \arrow[r, "\mathrm{id}"] & U
\end{tikzcd}$$

$$\begin{tikzcd}
\boxed{(U_i \cap U_j) \times F} \arrow[r, "\phi_j^{-1}"] \arrow[rr, bend right, "\phi_{ij} = \phi_i \circ \phi_j^{-1}"'] & \pi^{-1}(U_i \cap U_j) \arrow[r, "\phi_i"] & \boxed{(U_i \cap U_j) \times F}
\end{tikzcd}$$

Structure groups

Note for vect. bundles (def. #2) the transition functions (restricted to fibres)

$$(a,v) \in \{a\} \times \mathbb{R}^n \mapsto \phi_{ij}(a,v) \in \{a\} \times \mathbb{R}^n \longrightarrow \phi_{ij}(a,v) = (a, \dots)$$

form a group. Since they are linear, there are smooth maps

since $(a,v) \mapsto \phi_{ij}(a,v)$ is linear, it must be

$$\phi_{ij}(a,v) = (a, Av) \qquad A \text{ is an invertible matrix}$$

if $L : \mathbb{R}^n \to \mathbb{R}^n$ linear then $\exists A \in M_n(\mathbb{R})$ s.t. $L(v) = Av$

Note that A depends on a

we call $t_{ij}(a)$ the matrix s.t. $\phi_{ij}(a,v) = (a, t_{ij}(a)v)$

$$\underbrace{t_{ij} : U_i \cap U_j \to GL(n,\mathbb{R})}_{\to \text{ move these into fundamental objects}} \quad \text{smooth because } \phi_{ij} \text{ is smooth}$$

Def (G-atlas) Let $\pi : E \to M$ be a fibre bundle with fibre F.

Suppose there is a smooth left action of a Lie group G on F

$$(g,f) \in G \times F \mapsto \underbrace{g \triangleright f}_{g \text{ acting on } f} \in F \quad \text{smooth} \qquad \begin{cases} e \triangleright f = f & e \text{ identity of } G \\ g \triangleright (h \triangleright f) = (gh) \triangleright f \end{cases}$$

Then we call a G-atlas for E a collection of local trivialisations $\{(U_i, \phi_i)\}_{i \in I}$ s.t. the transition functions ϕ_{ij} satisfy

$$\phi_{ij}(u,f) = (u, t_{ij}(u) \triangleright f) \qquad \text{where } t_{ij} : U_i \cap U_j \to G \text{ are smooth functions}$$

we know that $\phi_{ij}(u,f) = (u, A_{ij}(a,f))$

assuming that $A_{ij}(a,f) \in F = g \triangleright f \in F$

$\rightarrow$ call this $t_{ij}(u)$

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t_{ij} are called <u>clutching functions</u>

G is called structure group of E.

ex: vector bundle has structure group $GL(n,\mathbb{R})$ acting linearly on fibre $F = \mathbb{R}^n$

$\boxed{\text{Properties of } t_{ij}}$

$t_{ii}(u) = e \quad \forall u$ $\leftarrow$ $\phi_{ii} = id$

$t_{ji}(u) = t_{ij}(u)^{-1}$ $\leftarrow$ $\phi_{ij} = \phi_{ji}^{-1}$

$t_{ij}(u) = \underline{t_{ik}(u)\, t_{kj}(u)}$ (cocycle property)

not a sum on k $\leftarrow$ $\phi_{ik} \circ \phi_{kj} = \phi_{ij}$

$\boxed{\text{Principal bundle}}$ a principal G-bundle is a fibre bundle $\pi: P \to M$ with fibre $F = G$ and structure group G, where the left action of G on F is left multiplication: $g \triangleright h = gh$

<u>Prop:</u> if $\pi: P \to M$ is a principal G-bundle with G-atlas $\{(U_i, \phi_i)\}_{i \in I}$ we can define a <u>right</u> action of G on P

if $p \in U_i$ then $p \triangleleft g = \phi_i^{-1}(\underbrace{\phi_i(p) \triangleleft g}_{=?})$

$\phi_i(p) \in U_i \times G$ if $(u,h) \in U_i \times G$ $\quad (u,h) \triangleleft g = (u, hg)$

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action satisfies:

- it's well-defined (does not depend on choice of ϕ_i)
- it preserves fibres: if $p \in F_a$ then $p \triangleleft g \in F_a$

- it's transitive on each fibre: for each $p, q \in F_a$ there is $g \in G$ s.t $q = p \triangleleft g$

- it's free: $p \triangleleft g = p$ for some $p \in P$ only if $g = e$

- the trivialisations are G-equivariant: $\phi_i(p \triangleleft g) = \phi_i(p) \triangleleft g$